Project Report: Flight Booking Price Prediction

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1. What is the Project?

This project is a machine learning solution to predict airline ticket prices based on features from historical flight booking datasets. The aim is to help users and businesses make *informed decisions* about flight bookings by forecasting prices using various input parameters such as airline, origin, destination, timings, stops, duration, and fare class.

2. Why was this Project Developed?

- **For Passengers:** To enable booking at optimal times and secure cheaper tickets.
- **For Airlines:** To understand market pricing trends and optimize revenue management.
- **For Educational Purposes:** To demonstrate the use of machine learning regression models on real-world tabular data and perform feature engineering, data cleaning, and model evaluation.

3. Algorithm Used

Most state-of-the-art flight price prediction projects, including those with similar structure and objectives, use:

- **Random Forest Regressor**: For robust, high-quality predictions due to its ability to capture complex, nonlinear relationships and feature importance.
- Sometimes, additional models like Linear Regression, Decision Trees, Lasso, Ridge Regression, or XGBoost are benchmarked.

4. Steps in the Project (What Was Done)

1. **Data Collection:** A CSV dataset (Flight_Booking.csv) containing flight details and prices.
2. **Exploratory Data Analysis:** Visualization and understanding of key variables influencing price.
3. **Data Preprocessing:** Handling missing values, encoding categorical features, and dropping irrelevant columns.
4. **Feature Selection:** Selecting variables with most predictive power (e.g., airline, stops, route, timings).
5. **Model Development:** Training models—Random Forest usually best for tabular flight data.
6. **Evaluation:** Using metrics like R-squared (R^2), Mean Squared Error (MSE), or Mean Absolute Percentage Error (MAPE) to assess performance.
7. **Prediction and Output:** Model predicts prices given user features.
8. **Result Analysis & Visualization:** Predicted vs. actual prices, residual analysis, and feature importances.
9. **Conclusion and Recommendation.**

5. Uses

- **Travel Agencies:** Real-time integration into booking sites to suggest optimal purchase times.
- **Airline Revenue Management:** Pricing strategy optimization.
- **Consumers:** Apps to notify when to book at lowest prices.
- **Researchers & Students:** Template for working with regression tasks and tabular data.

6. Accuracy & Results

- Best performing models (esp. Random Forest) reach **high R^2 scores, often >0.95** on test data, meaning they can explain more than 95% of the variance in ticket price based on the features provided.
- For example, similar projects report:
    - **R^2 >0.98** for Random Forest Regressor.
    - Other models (XGBoost, Extra Trees) also show excellent results with accuracies of 90–95%.
- **Visual outputs:** Plots compare true and predicted prices, and bar graphs show feature importances.

7. Output

- A trained ML model that accepts flight and booking details as input and returns a predicted fare.
- Often accompanied by:
    - Visualizations of price patterns (e.g. by day, airline, seasonality).
    - Plots of predicted vs. actual prices for model assessment.
    - Feature importance lists.

8. Conclusion

- **Random Forest** models excel at flight price prediction, achieving high accuracy on real-world data.
- Accurate predictions can save money for travelers and inform airline pricing strategies.
- Data cleaning, feature engineering, and algorithm selection are crucial for performance.